Virtual University of Pakistan

BT403

Agriculture Biotechnology

Final term past & current papers solved

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Mcqs

1. **Morphological markers** are those traits that are scored visually
2. A 2007 systematic review found that "most studies on non-Hodgkin lymphoma and **leukemia** showed positive associations with pesticide exposure" and thus concluded that cosmetic use of pesticides should be decreased.
3. Azolla is a free-floating water **fern** that floats in water and fixes atmospheric nitrogen in association with nitrogen fixing blue green alga Anabaenaazollae

4. A relative humidity of **50–60%** is maintained in the growth chambers.
5. Culture medium is sterilized in an autoclave for **20** min at 121o C at 15 psi (105 kPa).
6. At times of adverse environmental conditions, stress signals combined with **abscisic acid** (ABA) make the apoplast a prominent site for H2O2 production
7. Carotenoids belong to family of **lipophilic** antioxidants which are localized in the plastids of both photosynthetic and non-photosynthetic plant tissue
8. Azospirillum- It belongs to bacteria and is known to fix the considerable quantity of **nitrogen**
9. Diethyl aminoethyl dextran (DEAE-dextran) is a soluble poly cationic **carbohydrate** that promotes interactions between DNA and endo cytotic machinery of the cell.

All given waly 5 thy un me sb ka all given hi right ans tha

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What is synthetic seed.

• Synthetic seed can be defined as the artificial encapsulation of somatic embryo, shoot, bud, or aggregates of cells or any tissue which has the ability to form a plant in vitro.
• The use of synthetic seeds as an improvement on more traditional micropropagation protocols in facultatively propagated crops may, in the long term, have a cost saving, as the labour intensive step of transferring plants from in vitro to soil/field conditions may be overcome.

How gm can be used in medicine.

• These can be produced cheaper and easier some are: insulin, thyroid hormones and the Hepatitis B vaccine
• GM Bacteria‟s have been particularly important in producing large amounts of pure human proteins for use in medicine like clotting factors for hemophilia and human growth hormones to treat
dwarfism

What is biofertilizer Advantages of biofertilizer

• A biofertilizer (also bio-fertilizer) is a substance which contains living microorganisms which, when applied to seeds, plant surfaces, or soil, colonize the rhizosphere or the interior of the plant and promotes growth by increasing the supply or availability of primary nutrients to the host plant.
• Biofertilizers add nutrients through the natural processes of nitrogen fixation, solubilizing phosphorus, and stimulating plant growth through the synthesis of growth-promoting substances.

Advantages:

(i) They increase the yield of plants by 15-35%.
(ii) Bio-fertilizers are effective even under semi-arid conditions,
(iii) Farmers can prepare the inoculum themselves,
(iv) They improve soil texture,
(v) Bio-fertilizers do not allow pathogens to flourish,
(vi) They produce vitamins and growth promoting bio-chemical's,
(vii) They are non-polluting

What is agrobacterium transformation

• Agrobacterium tumefaciens is a soil-borne, Gram-negative bacterium. It is rod shaped and motile, and belongs to the bacterial family of Rhizobiaceae.
• A. tumefaciens is a phytopathogen, and is treated as the nature's most effective plant genetic engineer.

Advantages of PEG mediated transformation

1. A large number of protoplasts can be simultaneously transformed. 2. This technique can be successfully used for a wide range of plant species. **How plant defent against pathogens.** This defense device developed very rapidly in response to pathogen attacks.

• They have been linked to a self sealing car tyre and thought to make the host cell wall more defensive against pathogen penetration.

• Extensin has been considered such a cell wall protein which is subjected to cross linking.

Skoog wla b tha ek 5 marks ka Ethics long. Food med. Envi

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Write down the mineral which are added in basal media? 2 marks

What does stress with reference of plants?2

• water-logging
• drought

• lihigh or low temperatures
• excessive soil salinity
• inadequate mineral in the soil
• too much or too little light

Write down the type of biopesticides?3

1. Microbial pesticides
2. Plant-incorporated-protectants (PIPs)
3. Biochemical pesticides
4. Botanical pesticides
5. Biotic agents (parasitoids and predators)

What are the method by which plants become stress tolerant?3

• Adaptation: heritable modifications to increase fitness
• CAM plants' morphological and physiological adaptations to low H2O Environment
• Acclimation: non heritable physiological and biochemical gene expression
• Cold hardening induced by gradual exposure to chilling temps, e.g. cold-hardy Plants

How doed stock solution are prepared for tissue culture media?3

Preparation of stock solutions: Since it is a time - consuming and tedious process to weigh the necessary products each time a medium is prepared, concentrated solutions of the desired composition of a medium are used which one dilutes adequately. These concentrated solutions are called stock solutions. Simple stock solutions comprise only one constituent at a time. Complex stock solutions comprise several chemicals. Stock solutions of macro and micronutrients, vitamins and growth regulators are prepared in distilled or high purity demineralized water. Chemicals should be of the highest grade.

i. Macronutrient stock solution(s): Usually, the stock solution of macronutrients is prepared as 10x. Dissolve all the macronutrients one by one except (CaCl2 for macronutrient stock solution. The stock solution of CaCl2 should be prepared separately. Another way is to dissolve the different macronutrients one after the other and CaCl2 is dissolved separately and later added to the rest of the stock solution in order to avoid precipitation.

ii. Micronutrient stock solution: A stock solution of all the micronutrients with 100x is generally prepared. Since copper and cobalt are required in very small quantities, it is preferable to first make a separate stock solution of these two salts (100x) and then an appropriate volume can be pipetted and put into the main micronutrient stock solution. These nutrient solutions can be dispensed in plastic bags with zipper seals and stored frozen (e.g. 10x macronutrient solution is dispensed into a bag containing 100ml of solution to prepare 1 litre medium). .

What are molecular markers?5

• A molecular marker is a molecule contained within a sample taken from an organism (Biochemical molecules) • The first biochemical molecular markers used were the protein based markers.

• One of the earliest protein based markers to be used was Isozyme.

• These are different forms of an enzyme exhibiting the same catalytic activity but differing in charge and electrophoretic mobility.

Types of Molecular markers

Morphological markers:
• Morphological markers are those traits that are scored visually, or morphological markers are those genetic markers whose inheritance can be followed with the naked eye.

Biochemical molecular marker

• A molecular marker is a molecule contained within a sample taken from an organism (Biochemical molecules)
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• One of the earliest protein based markers to be used was Isozyme.
• These are different forms of an enzyme exhibiting the same catalytic activity but differing in charge and electrophoretic mobility.

DNA based markers
• The sequence of nucleotides in DNA of an individual is unique and thus determines its identity.
• The ultimate difference between individuals lies in the nucleotide sequence of their DNA.
• These can be used to diagnose the presence of the gene without having to wait for gene effect to be seen.

What is high lysine corn 🌽?5

Under the heading of 'High-lysine com', we have described how cereals rich in certain essential amino acids such as lysine, methionine and tryptophan can be developed by genetic engineering.

• Likewise, rice is being modified into Golden rice by Prof. Inge Potrykus and Dr. Peter Beyer.
• This is done so that vitamin A potential is maintained even after the husks are removed, a procedure adopted to allow for storage since the husks become rancid.

• This change may improve health of millions of people throughout the world.

What are ethical aspects of agricultural biotechnology with respect of food medicine and environment?10

Food

• Agriculture has always depended on plant and animal breeding and modern biotechnology provides new possibilities.
• Public perceptions of agro-food biotechnology are more critical than of its applications in healthcare. This probably results from the cultural and symbolic functions of food together with most people's relative ignorance about modern agriculture and food production.
• The most important areas where biotechnology can provide benefits for European consumers can be in improved price, quality and nutritional value of foods.

• Ethical considerations of agrofood biotechnology relate to the environment, biodiversity, sustainability, animal welfare and its socio-economic impacts.
• Consumers rights concerning biotechnological food products relate to the rights to health from safe foods, to be informed and to choose genetically engineered products or not.
• It is crucial that balanced information is provided to the public.
• Communication strategies should bring together the scientific, industrial and general communities to promote openness, dialogue and mutual understanding.

Environment

• The current environmental problems that arise from agriculture stem from modern, intensive agricultural practices and not from the use of genetically modified crops, as the latter are only currently being introduced into European agriculture.
• The use of biotechnology may either exacerbate or ameliorate these effects depending on how it is applied.
• It is therefore important, before applying new biotechnology, to consider the precautionary principle, the need for sustainable development and the need to maintain and possibly enhance agriculturally-important biodiversity.

Medicine

• Biotechnology offers many opportunities for the production of medicines, vaccines and other medical products using agricultural sources for further improvement of human and animal health.
• New products can be developed using this technology, or the production of already existing products made more cost-effective.
• The disadvantages of the use of biotechnology must not be overlooked: although safety regulations do exist unforeseen and unwanted consequences may still occur.

Cell suspension culture. 5marks

 A cell suspension culture consists of cell aggregates dispersed and growing in a moving liquid media.
 It is normally initiated by transferring pieces of undifferentiated and friable callus to a liquid medium, which is continuously agitated by a suitable device.
 Suspension cultures can also be started from sterile seedlings or imbibed embryos or leaves by the mechanical method.

 Equipments:
Shaker
Centrifuge
Microscope
Haemocytometer

 Reagents and other requirements
Sterilants – alcohol, HgCl2, sodium hypochlorite All given
Nutrient medium reagents -MS basic salts and vitamins
Growth regulators - 2, 4-D
Seedlings of *Phaseolus mungo*

 Media
 MS + (1.5mg/l) 2, 4-D

Productivity losses due to stress. 5marks

Productivity losses due to stress

• Loss due to diseases range from 20 to 30 %, in case of severe infection, total crop may be lost.
• Estimated global loss due to insect pests in potential yields of all crops is ~14%.
• Losses due to insect pests ranges from 10 to 20 %
• Abiotic stresses reduce average yield of crops by upto50% .
• Most of the agricultural area is rain fed and crops in these areas invariably experience droughts of different magnitudes.
• Annually about 42% of the crop productivity is lost owing to various

abiotic stress factors.

Ethical aspects of agricultural biotechnology in food, medicine and environment. 10 marks

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Artificial Seed Types of Invitro cell culture
Types of biopesticides

Principal of Anthur culture experiment

 Principle:

 Haploids refer to those plants which possess a gametophytic number of chromosomes in their sporophytes. Haploids may be grouped into two broad categories:

 (a) monoploids which possess half the number of chromosomes from a diploid species.

 (b) Polyhaploids which possess half the number of chromosomes from a polyploidy species.

 Haploid production through anther culture has been referred to as androgenesis while gynogenesis is the production of haploid plants from ovary or ovule culture where the female gamete or gametophyte is triggered to sporophytic development.

Differ b/w gm and selective breeding

Principle of electroporation?

Ans: • Electroporation is another method of promoting competence. • In the method the cells are briefly shocked with an electric field of 10-20 kV/cm that creates holes in the cell membrane through which the plasmid DNA enters.
• This method is ready to the uptake of large plasmid DNA. • After the electric shock the holes are rapidly closed by the cell's membrane-repair mechanisms. • The efficiency with which a competent culture can take up exogenous DNA and express its genes is known as Transformation efficiency

Two source of tolerance of herbicides

Ans: • (1) Over-production of the target enzyme and
• (2) Mutations resulting in enzymes that are less sensitive to the inhibitor (usually due to a lower affinity of the enzyme for the inhibitor).

Treatments for skin pesticides?

Ans: • Rinse well with cool water. • Do not remove anything stuck to the burn. • Do not apply lotions, fats, or butter. • Do not break blisters. • Do not remove loose skin. • Cover the area with a sterile dressing, if available.

Gene splicing?

Ans: With this GM technique, biotechnologists can modify DNA, and then insert it into target host cells to allow for genes and resulting traits to be modified.

• An enzyme is then used to fuse the newly added gene into the chromosome.

Multipartite viruses?
Ans: The genome in these viruses is divided into small RNAs which may be in the same particle or different particles, e.g. brome mosaic virus (BMV). HMV contains four RNAs divided between three particles. Plant RNA viruses, in general, are characterized by high level of gene expression, good efficiency to infect cells and spread to different tissues. But the major limitation to use them as vectors is the difficulty of joining RNA molecules in vitro.

Competent cells?
Ans: Competent Cells are more likely to incorporate foreign DNA if their cell walls are altered so that DNA can pass through more easily. • Such cells are said to be competent
Types of competence
• Natural competence • Artificial competence (further details in topic 26)

Lettuce why used instead of tobacco for human growing plants
Ans: Most of the plant is leaf tissue and this tissue contains the greatest number of plastids per cell • Unlike tobacco, lettuce has no toxic alkaloids that need to be removed - low purification and downstream processing costs • Lettuce is a relevant human foodstuff that can be consumed without cooking
how Gm adds more nutrition value to plants

Ans: • The GMO method can put in added nutritional value to crops that lack necessary vitamins and minerals. • Considering that there are places in the world relying on rice or corn as their daily staple, plant genes may be added to these crops to increase their nutritional value.
• This would help malnourished populations receive more nutrients from their diet.

tissue culture types (long)
Ans: • Types of in vitro cultures 1. Callus Cultures 2. Cell Suspension Culture 3. Anther/Microspore Culture 4. Protoplast Culture 5. Embryo Culture 6. Meristem Culture
Callus Culture
• An unorganized mass of thin-walled parenchyma cells
• Formation of wound callus is observed in all groups of living organisms
• In Plant tissue culture, callus is produced by cultivation of plant tissue on nutrient media under in vitro conditions
• Presence of growth hormones in the culture media promotes callus formation and proliferation
Cell Suspension Cultures
• Cell suspension cultures are rapidly dividing suspensions of cells grown in liquid medium
• Grow more rapidly than callus cultures and more ready to culture manipulations
• Comprised of cell aggregates and dispersed single cells
Anther/Microspore Culture
Anthers or pollens can be cultured on a suitable medium containing sucrose (usually 2%), iron, vitamins, hormones etc.
• The hormonal component of the medium is important for initiation of growth.
• Usually to the culture medium auxin, cytokinin etc. are added either singly or in various combinations.
• Low concentration of auxin stimulates callus formation. In a medium supplemented with auxin embryoid formation usually occurs at a faster rate as observed in anther culture of Datura.
• Anthers cultured on a medium containing coconut milk or kinetin develop embryoids which later form haploid plantlets.
• Callus is formed from pollen grains on a medium supplemented with yeast extract or casein hydrolysate.
Protoplasts culture
• Protoplasts are naked plant cells without the cell wall, but they possess plasma membrane and all other cellular components. • They represent the functional plant cells but for the lack of the barrier, cell wall. Protoplasts of different species can be fused to generate a hybrid and this process is referred to as somatic hybridization (or protoplast fusion). • Cybridization is the phenomenon of fusion of a normal protoplast with an enucleated (without nucleus) protoplast that results in the formation of a cybrid or cytoplast (cytoplasmic hybrids).
Embryo Culture
The embryo of different developmental stages, formed within the female gametophyte through sexual process, can be isolated aseptically from the bulk of maternal tissues of ovule, seed or capsule and cultured in vitro under aseptic and controlled physical conditions in glass vials containing nutrient solid or liquid medium to grow directly into plantlet
Meristem Culture

• Meristem culture is the in vitro culture of a generally shiny special dome-like structure measuring less than 0.1 mm in length and only one or two pairs of the youngest leaf primordia, most often excised from the shoot apex.

pesticide poisoning types

Ans: Types of pesticide poisoning:

• There are two types of pesticide poisoning:

Acute poisoning

• This happens when someone has been exposed to a high dose of pesticide.

• This could occur when the pesticide is being mixed, for example, or if a hose breaks drenching the person or bystanders with liquid pesticide solution.

• Another example might be accidental ingestion of a pesticide, such as a child swallowing the chemical.

Chronic poisoning

• This results from a person being exposed to a small amount of pesticide on many occasions over a long period of time.

• Chronic poisoning may happen when the operator repeatedly uses pesticide improperly, especially if they does not wear protective clothing and equipment or wears protective clothing which is not clean or is worn out, like wearing cracked or torn gloves.

Meristem culture principle

Ans: • Meristem culture is the in vitro culture of a generally shiny special dome-like structure measuring less than 0.1 mm in length and only one or two pairs of the youngest leaf primordia, most often excised from the shoot apex.

Principle

• The excised shoot tips and meristem can be cultured aseptically on agar solidified simple nutrient medium or on paper bridges dipping into liquid medium and under the appropriate condition will grow out directly into a small leafy shoot or multiple shoots.

• Alternatively the meristem may form a small callus at its cut case on which a large number of shoot primordia will develop.

losses of plants due to diseases

Ans: The most prominent biotic stress that the plants experience is due to disease outbreaks. • Damages lead by insect interference is also gaining a pick position regarding the biotic stress of plants. • A plant Disease may be defined as the series of invisible and visible responses of plant cells or tissues to a pathogenic microorganism or an environmental factor that results in adverse changes in form, function & integrity of plants and may lead to partial impairment or death of the plant or its parts

Giberlliene advantages

Ans: Gibberellins, although found in all plants and fungi, are unevenly distributed. The sites of synthesis are very young, unopened leaves, active buds, root tips and embryos. There are various gibberellin compounds with a similar gibbane nucleus but differ in the quality and position of the substituents of the nucleus. Gibberellic acid (GA3) is the most widely used compound. Gibberellins are normally used in plant regeneration.

GA3 is essential for meristem culture of some species. In general, gibberellins induce elongation of internodes and the growth of meristems or buds in vitro.

In its absence, the culture appears globular, due to the accumulation of nodes. Gibberellins usually inhibit adventitious root as well as shoot formation.
During organogenesis, gibberellins are antagonistic. They seem to oppose the

phenomenon of dedifferentiation.
Thus, in in vitro cultures, they cannot be used for this purpose, but can be utilized for explants already organized (meristems, apices, buds).

Name of antioxidant enzymes

- Anionic peroxidases Chilling, high CO2
- Ascorbate peroxidase Drought, high CO2, high light intensity, ozone, paraquat
- Catalase Chilling
- Glutathione Chilling drought, irradiation, heat stress, high CO2, ozone, SO2
- Glutatione reductase Chilling, drought, high CO2, ozone, paraquat
- Polyamines

Lab animals tested with gm food

Lab animals tested with gm foods
• Stunted growth
• Impaired immune systems
• Bleeding stomachs
• Impaired blood cell development
• Misshapen cell structures in the liver, pancreas, and testicles
• Altered gene expression and cell metabolism
• Their offspring have a lower

• chance of survival
• Higher blood sugar
• Enlarged livers, pancreases
• Inflamed kidneys
• Less developed brains and testicles
• Intestines reduced digestive enzymes
• The animal‟s life spans are shortened
• Inflamed lung tissue

Sequestering of herbicides resistance

• Some plants are capable of restricting the movement of foreign compounds (herbicides) within their cells or tissues to prevent the compounds from causing harmful effects.
• In this case, an herbicide may be inactivated either through binding (such as to a plant sugar molecule) or removed from metabolically active regions of the cell to inactive regions, the cell wall, for example, where it exerts no effect.

Effects of Heat Stress

• Dehydration.
• Yellowing of plants and leaves; chlorophyll deteriorates.
• Withered leaves
• Sunburn; reddish-purple tint or freckles on leaves and pseudo bulbs, leaf tip and root may turn brown.
• Alteration of gene expression is the major cause of heat stress.
• Leathery leaves; damage on cellular level.
• In response to high temperature all organisms, including plants, synthesize a set of proteins called as heat shock proteins (HSPs) The induction of HSPs at permissive temperatures have been associated with the acquisition of thermotolerance to withstand the stress

Roundup Ready crops

• 'Roundup Ready' soya:
most widely grown
GM crop variety
• tolerates applications
of Monsanto's 'broad
spectrum' glyphosate
herbicide
• Roundup destroys all
other plants.

• Italian mouse trial:
Roundup Ready soya
affects key body organs
• FSA human feeding trial:
entire transgenic gene in
GM soy survives the
passage through the
stomach and small
intestine, though not
through the colon
• Roundup residues cause
cell damage

techniques of plant tissue culture

• Culture vessels
• Culture medium
• Sterilization
• Inoculation
• Incubation
• Induction of callus

• Morphogenesis = Organogenesis Embryogenesis
• Hardening

Bio Compost:
• It is one of the Eco-friendly product composed of waste material released from sugar industries which are decomposed.
• It is magnified with human-friendly bacteria, fungi, and various plants.

Culture of protoplasts
• Protoplasts cultured in suitable nutrient media first generate a new cell wall
• The formation of a complete cell with a wall is followed by an increase in size, number of cell organelles, and induction of cell division
• The first cell division may occur within 2 to 7 days of culture
• Resulting in small clumps of cell, also known as micro colony, within 1 to 3 weeks
Importance of Protoplast Culture
• The protoplast in culture can be regenerated into a whole plant.
• Hybrids can be developed from protoplast fusion.
• It is easy to perform single cell cloning with protoplasts.
• Genetic transformations can be achieved through genetic engineering of protoplast DNA.
• Protoplasts are excellent materials for ultra-structural studies.

Constant flow system in microinjection?

Ans: In the constant flow system the amount of sample injected is determined by the duration for which needle remains in the cell. • The constant flow system is relatively simple and inexpensive but outdated.

Role of flavonoids in plants

Ans: They have diverse roles in providing pigmentation in flowers, fruits and seeds involved in plant fertility and germination of pollen and defense against plant pathogens.

• Flavonoids have been considered as a secondary ROS scavenging system in plants experiencing damage to the photosynthetic apparatus, due to the excess excitation energy.

• They also have a role in scavenging 1O2 and alleviate the d

Germplasm Preservation

• Germplasm is living tissue from which new plants can be grown. It can be a seed or another plant part a leaf, a piece of stem, pollen or even just a few cells that can be turned into a whole plant. • Germplasm contains the information for a species' genetic makeup, a valuable natural resource of plant diversity.

Genetic marker

Ans: A genetic marker is a gene or DNA sequence with a known location on a chromosome

• It can be used to identify individuals or species.

• It can be described as a variation (which may arise due to mutation or alteration in the genomic loci) that can be observed.

6 symptoms of sever poisoning

Ans: • vomiting • convulsions • loss of reflexes • unconsciousness • inability to breathe

• fever • muscle twitching • thirst • constriction of eye pupils (eye pupils become small) • increased rate of breathing

How viruses transfer from one plant to another

Ans: Spread

• Viruses can be spread from plant to plant by several means. Some of these would include transmission from the parent plant to an offspring through the genetic structure of the plants. • Other ways in which viruses can be transmitted are through vegetative propagation, grafting and budding, seed transmission and mechanical spread by insects and man.